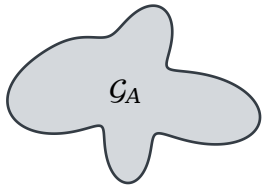
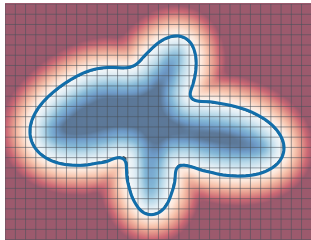


(a) Irregular geometry



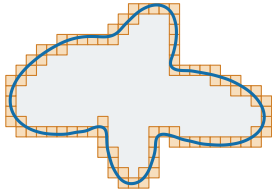
Original boundary

(b) Patch-wise SDF



Fit ϕ_A ; certify level l_A

(c) Reuse surface patches



$\mathcal{S}_A = \{\phi_A = l_A\}$

(d) Finite control constraints

Bernstein + Taylor bounds

$$L_{kj}(\mathbf{u}, \mathbf{a}) \geq 0$$

$j = 1, \dots, 4$ per patch



$$\dot{h}(\mathbf{x}) + \gamma h(\mathbf{x}) \geq 0$$

$$\forall \mathbf{x} \in \mathcal{S}_A \cap \mathcal{Q}_k$$

One spline representation: native patches certify the entire continuous surface.